Arrays_Doubt

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(i) next permutation
(ii) maximum circular sum
(iii) next greater
(iv) calculate the sum
(v) maximum sum path in 2 Array

[3, 4, 1, 9, 8, 7, 6, 4, 3, 2, 1]

0 1 2 3 4 5 6 7 8 9 10

[3 4 1 9 8 7 6 3 3 2 1]

3 4 1 4 1 2 3 3 7 7 8

(i) P
(ii) Q
(iii) P & Q
(iv) P+1 to n-1 Reverse

[1 -2 3 -4]

[4 3 -7 1 -4 5 -1 3] linear sum / circular sum

[-2, -3, 7 -1, 4, -2] → Kadane

4+3 + (-7+1-4) + 5 -1 + 3 + 10 = 10

[4 3 -7 1 -4 5 -1 3]

[-4 -3 7 -1 4 -5 +1 -3]

total + 1

max

min

```java
public static int Circular_Subarray(int[] nums) {
    int linear_sum = Kadanes_Algorithm(nums); → -1
    int total_sum = 0;
    for (int i = 0; i < nums.length; i++) {
        total_sum += nums[i];
        nums[i] = -1 * nums[i];
    }
    int middle_sum = Kadanes_Algorithm(nums);
    int Circular_sum = total_sum + middle_sum;
    return Math.max(linear_sum, Circular_sum);
}
```

(-1, 0) → 0

[-2 -1 -3] = -6
[2 1 3] = 6
] = 0

(1) cab — 160m

(1) rickshaw — n — 1 to n

1 3 7 19

2 3

2 5

4 4 4

2×(2×1, 3) = 2

5×(5×1, 3) = 3 → (5, 7) = 8

(4×1, 3) = 3

(3×1, 3) = 3 = 9 → 7

min [12, 14] = 12

1 5 7 = 13 + 18 + 27 + 64

8 8

2 3 7 10 12 15 30 34

1 5 7 8 10 15 16 19

S1-i ✓ 10 27

S2-j ✓ 18 15

```java
public static int Maximum_Score(int[] arr1, int[] arr2) {
        int s1 = 0, s2 = 0, i = 0, j = 0;
        long sum = 0;
        int mod = 1000_000_007;
        while (i < arr1.length && j < arr2.length) {
            if (arr1[i] < arr2[j]) {
                i++;
            } else if (arr1[i] > arr2[j]) {
                j++;
            } else {
                // s1 to i
                long sum1=sum_of_Array(arr1, s1, i);12
                // s2 to j
                long sum2=sum_of_Array(arr2, s2, j);13
                sum+=Math.max(sum1, sum2);

            }

        }
    }
```

i++, j++
s1=i, s2=j

sum = 13 + 18 + 27

2 3 7 10 12 15 30 34

1 5 7 8 10 15 16 19

5

1 2 3 4 5

2

1

2

| 1 | 2 | 3 | 4 | 5 |

6 3 5 7 9

| 6 | 3 | 5 | 7 | 9 |

| 13 | 12 | 11 | 10 | 14 |

| 1 | 2 | 3 | 4 | 5 | 6 |

| 6 | 8 | 4 | 6 | 8 | 10 |

0 1 2 3 4 5

n[i] = a[i] + a[n[i]]

x = 2

k = 4

j = i - x = 0 - 2 = -2

j = 1 - 2 = -1 + 4 (=3)

j = 2 - 2 = 0

j = 3 - 2 = 1

j = 4 - 2 = 2

j = 5 - 2 = 3